



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN MECHANICAL ENGINEERING,
BACHELOR OF SCIENCE IN CIVIL ENGINEERING AND
BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONIC ENGINEERING
ECU 203: TRANSFORM METHODS

DATE: Monday, 1st July 2019

TIME: 8.00a.m-10.00a.m

INSTRUCTIONS

Answer question ONE and any other TWO

Question One (30 marks)

1. (a) i. Define a Gamma function

(2 marks)

ii. Evaluate the Integral

$\int_0^{\infty} \sqrt{y} e^{-t} dt$ using definition of (a) (i) above

(4 marks)

(b) Use Power Series method to solve the differential equation

$y^{(4)} + (x^2)y = 0$ near ordinary point $x_0 = 0$

(6 marks)

(c) Solve the differential equation

$y^{(4)} - 7y'' + 12y = e^x$ $y(0) = 1$, $y'(0) = 2$ using laplace transform (6marks)

(d) Given that $F(s) = \frac{s+2}{s^2-4s+13}$ find $f(t)$

(6 marks)

(e) Determine the Fourier Series of a Periodic function of the voltage from a square wave generator which is of the form

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(6marks)

$$V(t) = \int_0^t \begin{cases} -4 & -4 < t < 0 \\ 0 & 0 < t < 4 \end{cases} \text{ with a period of 8ms}$$

Question TWO (20 marks)

2. (a) Define the terms below with respect to the 2nd order linear differential equation

(i) An Analytic Function

Differentiable at x_0

There exist a neighbourhood V of x_0 such that f is derivable at every point

(2marks)

(ii) A Singular Point

$$y'' + p(x)y' + q(x)y = 0$$

If $p(x)$ and $q(x)$ are analytical at x_0 .

(2marks)

(iii) An Ordinary Point

(2marks)

not analytical

(b) Determine the Particular Polynomial solution to the Legendree equation

$$(1-x^2)y'' - 2xy' + 42y = 0$$

(Series solution not necessary)

(7marks)

(c)

Find the recurrence relation of the differential equation $y'' - xy' - 2y = 1 + x$ using Power Series near Ordinary Point $X_0 = 0$.

(7marks)

Question Three (20 marks)

3. (a) (i) Define a laplace Transform

(3marks)

(ii) Using the definition stated above in 3(a) (i) to find laplace transform of

$$f(t) = t^2 + e^{3t}$$

(6 marks)

(b) Find the 2nd order d.e using the Laplace Transform.

$$y'' + 9y = 6 \cos 3t \text{ for } y(0) = 2, \text{ and } y'(0) = 0$$

(6 marks)

Question Four (20 marks)

4. (a) Define the Fourier expansion Series

(4marks)

(b) State the difference between odd and even functions and list down the Properties of each one of them.

(6marks)

(c) Determine the Fourier Series for the function defined by

ECU 203

1st July 2019

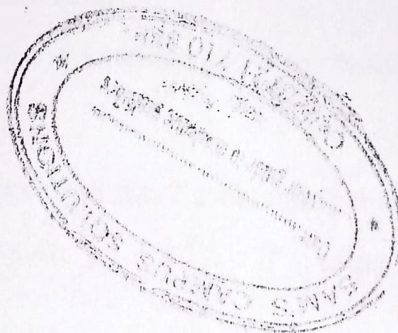
$$f(x) \int_0^1 \begin{matrix} \text{where } -3 < x < -1 \\ 5 \text{ where } -1 < x < 1 \\ 1 < x < 3 \end{matrix}$$

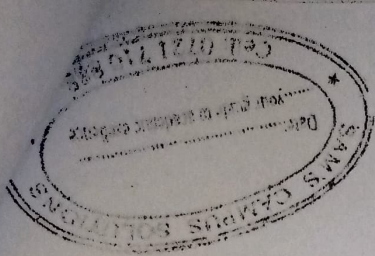
Period 6

Question Five (20 marks)

(10marks)

- 5(a) Find the general solution to the Bessel equation $x^2 y'' + xy' + (x^2 - 4)y = 0$ (5marks)
- (b) Expand $f(x) = x^3$ for $-\pi < x < \pi$ into Fourier Series (10marks)
- (c) Solve the differential equation $y' = y^2 + 1$ for $y(\pi/4) = 1$ (5 marks)





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UNIVERSITY EXAMINATIONS 2017/2018
SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (MECHANICAL ENGINEERING),
BACHELOR OF SCIENCE (CIVIL ENGINEERING), AND
BACHELOR OF SCIENCE ELECTRICAL AND ELECTRONICS ENGINEERING)
ECU 203: ENGINEERING MATHEMATICS VIII

DATE: Thursday 30th August 2018

TIME: 11.00a.m-1.00p.m

INSTRUCTIONS:

Answer Question ONE and any other TWO

Question One

- (a) (i) Define a Gamma function. (4 marks)
(ii) Use the definition of gamma above to evaluate

$$\int_0^{\infty} \sqrt{x} e^{-x^3} dx \quad (4 \text{ marks})$$

- (b) Determine the recurrence formula for obtaining the co-efficients of general solution of the differential equation $y'' + x^2 y = 0$ using Power Series near ordinary point $x_0 = 0$. (4 marks)
(c) Find $\int x^3 \sin x dx$ using the New theorem of integration by parts. (4 marks)
(d) The voltage from a square wave generator is of the form

$$V(t) = \begin{cases} 0 & -4 < t < 0 \\ 10 & 0 < t < 4 \end{cases}$$

And has a period of 8ms. Find the Fourier Series for this Periodic function.

(7 marks)

- (e) Use Laplace Transforms to find the solution of the initial value problem.

$$y'' + 6y' + 8y = e^{3x} \quad \text{for } y(0) = \frac{52}{51} \quad y'(0) = \frac{35}{17} \quad (7 \text{ marks})$$

Question Two (20 marks)

- a) i) Define a Laplace transform.
 ii) Use definition stated in 2(a) (i) above to obtain the Laplace transform of $f(x) = x^2 + e^x$ (3 marks)
- b) Find the inverse Laplace Transform of:-
 (i) $F(s) = \frac{s+2}{s^2-4s+13}$ (5 marks)
 (ii) $F(s) = \frac{s+4}{s(s-2)(s^2+4)}$ (5 marks)
- c) Solve the IVP using Laplace Transform $y'' + 9y = 6 \cos 3t$ for $y(0) = 2$ and $y'(0) = 0$

Question Three (20 marks)

- a) Use Power Series method to solve the differential equation $y'' + xy' - 2y = 0$ for $y(0) = 1$ and $y'(0) = 0$ near ordinary point $x_0 = 0$. (8 marks)
- b) Determine the particular polynomial solution to the Legendree equation $(1-x^2)y'' - 2xy' + 42y = 0$.
 (Series solution not necessary) (6 marks)
- c) Find the general solution to the Bessel equation $x^2y'' + xy' + x^2 - 4 = 0$. (6 marks)

Question Four

- a) Define the Fourier expansion series (3 marks)
 b) State the conditions that Dirichlet found out to satisfy the convergence of Fourier series. (4 marks)
 c) Define the odd and even functions and state the properties of each function. (6 marks)
 d) Obtain the Fourier series for the function defined by
- $$f(x) = \begin{cases} 0 & \text{when } -2 < x < -1 \\ 5 & \text{when } -1 < x < 1 \\ 0 & \text{when } 1 < x < 2 \end{cases}$$
- (7 marks)

Question Five

- a) Solve the differential equation $y' = y^2 + 1$ subject to $y(0) = 1$. Using any method. (6 marks)
- b) Convert the differential equation below into a Laplace Transform and hence solve $3y'' + 7y' + 4y = 2e^{7x-2}$
 Given that $y(0) = 1$ and $y'(0) = 2$ (7 marks)
- c) Find the inverse Laplace Transform of:
 $\frac{2s^2 - 6s + 5}{(s-1)(s-2)(s-3)}$ (7 marks)

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KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2019/2020

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN MECHANICAL ENGINEERING,
BACHELOR OF SCIENCE IN CIVIL ENGINEERING AND
BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONIC ENGINEERING
ECU 203: ENGINEERING MATHEMATICS III (TRANSFORM METHODS)

DATE: Tuesday 26th January 2021

TIME: 8.00a.m-10.00a.m

INSTRUCTIONS

Answer QUESTION ONE (COMPULSORY) and ANY other TWO questions

QUESTION ONE (30 MARKS)

- Define the Gamma function and hence evaluate $\int_0^\infty y^2 e^{-2y^2} dy$ (6 marks)
- If $B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$, determine $B(2.5, 1.5)$ (3 marks)
- Given that $F(s) = \frac{1}{3-4s} + \frac{3-2s}{s^2+49}$, determine $f(t)$ (5 marks)
- Obtain the value of $P_3(x)$ using Rodrigues' formula (5 marks)
- Determine the Fourier series of a periodic function of the voltage from a square wave generator which is of the form $V(t) = \begin{cases} 0, & -4 < t < 0 \\ 10, & 0 < t < 4 \end{cases}$ with a period of 8m/s (5 marks)
- A first-order differential equation involving current i in a series R-L circuit is given by:
 $\frac{di}{dt} + 5i = \frac{E}{2}$ and $i = 0$ at time $t = 0$. Use Laplace transforms to solve for i when $E = 40e^{-3t}$ (6 marks)

QUESTION TWO (20 MARKS)

- Show that $\sin \theta \frac{d^2 y}{d\theta^2} + \cos \theta \frac{dy}{d\theta} + n(n+1) \sin \theta y = 0$ can be transformed into Legendre's equation by means of the substitution $x = \cos \theta$ (9 marks)
- Determine the half range Fourier sine series to represent $f(t) = t - t^2, 0 < t < 1$ (5 marks)

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c) Prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

(6 marks)

QUESTION THREE (20 MARKS)

- a) Determine whether $x = 0$ is an ordinary point, a regular singular point, or an irregular singular point of the differential equation $x^4 y'' + (x^2 \sin x) y' + (1 - \cos x) y = 0$ (4 marks)
- b) Given that $F(s) = \frac{6s^2 + 62s + 92}{(s+1)(s^2 + 10s + 21)}$, determine $f(t)$ (5 marks)
- c) Using the Beta function, show that $\int_0^2 \sqrt{x(2-x)} dx = \frac{\pi}{2}$ (5 marks)
- d) Determine the particular polynomial solution to the Legendre's equation $(1 - x^2) y'' - 2xy' + 42y = 0$ (series solution not necessary) (6 marks)

QUESTION FOUR (20 MARKS)

- a) Use Laplace transforms to solve the differential equation: $y'' - 2y' + 2y = 3e^t \cos t$, given $y(0) = 2$ and $y'(0) = 5$ (8 marks)
- b) Define the Laplace transform and hence determine the Laplace transform of the function $f(t) = t(t-1) - \sin 2t + e^{3t}$ (5 marks)
- c) Given that $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ and $J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$, using the identity;
- $$\frac{d}{dx} [x^{-p} J_p(x)] = -x^{-p} J_{p+1}(x)$$
- Show that; $J_{\frac{3}{2}}(x) = \sqrt{\frac{2}{\pi x}} \left[\frac{\sin x}{x} - \cos x \right]$ (7 marks)

QUESTION FIVE (20 MARKS)

- a) Evaluate the integral $\int_0^\infty u^{\frac{3}{2}} e^{-3u} du$ (5 marks)
- b) Determine the first four terms in each portion of the series solution around $x_0 = 0$ for the differential equation: $(x^2 + 1) y'' - 4xy' + 6y = 0$ (6 marks)
- c) Obtain the Fourier series expansion of: $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$
- Hence deduce that;
- $$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

(9 marks)

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UNIVERSITY EXAMINATIONS 2016/2017

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BACHELOR OF SCIENCE ELECTRICAL AND ELECTRONICS ENGINEERING)

ECU 203: ENGINEERING MATHEMATICS VIII

DATE: Monday, 12th June 2017

TIME: 2.00 p.m. - 4.00 p.m.

INSTRUCTIONS:

Answer Question ONE and ANY OTHER TWO.

QUESTION ONE (COMPULSORY) - 30 MARKS

1. (a) (i) Define a Laplace transform. (2 marks)
(ii) Using the definition in 1(a)(i) above, determine the Laplace transform of the function $f(t) = t^2 + e^{3t}$. (4 marks)
- (b) Determine the recurrence relation of the differential equation $y'' - xy' - 2y = 0$ using Power Series near ordinary point $x_0 = 0$. (5 marks)
- (c) Evaluate the following gamma function

$$\frac{6\sqrt{8/3}}{5\sqrt{2/3}} + \sqrt{-5/2}$$

(5 marks)

- (d) Given the function $f(x)$ as

$$f(x) = \begin{cases} 0 & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$$

and that $f(x + 2\pi) = f(x)$,

- (i) Sketch the function $f(x)$. (2 marks)
- (ii) State the period. (1 mark)
- (iii) Calculate the fourier co-efficients of $f(x)$. (2 marks)

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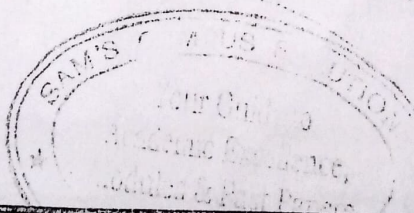
- (e) Determine a particular polynomial solution to the Legendree equation $(1 - x^2)y'' - 2xy' + 30y = 0$. (series solution not necessary) (4 marks)
- (f) Obtain the particular solution of the differential equation $2y'' + 5y' + 2y = e^{-2t}$ for $y(0) = 1$ and $y'(0) = 1$ using the Laplace Transform method. (5 marks)

QUESTION TWO (20 MARKS)

2. (a) State the Fourier Series of a function $f(t)$. (3 marks)
- (b) The periodic function of the voltage from a square wave generator is of the form
- $$V(t) = \begin{cases} 0 & -4 < t < 0 \\ 10 & 0 < t < 4 \end{cases}$$
- with a period of 8ms. Determine its Fourier Series. (8 marks)
- (c) (i) Differentiate between even and odd functions. (2 marks)
- (ii) Using (i) above determine whether the function $f(x) = x^3 \sin x$ is an odd or an even function. (3 marks)
- (iii) If $f(x)$ is an even function evaluate $\int_{-a}^a f(x) dx$. What can you say about the integral $\int_{-a}^a f(x) dx$ if $f(x)$ is an odd function? (4 marks)

QUESTION THREE (20 MARKS)

3. (a) (i) Define the gamma function. (3 marks)
- (ii) Use the definition in 3(a)(i) above to evaluate the following integral:
- $$\int_0^{\infty} \sqrt{y} e^{-y^3} dy + \int_0^{\infty} \sqrt{x} e^{-3\sqrt{x}} dx. \quad (7 \text{ marks})$$
- (b) The Beta function is defined as $B(x, y) = \frac{\int_0^1 x^y y^x}{\int_0^1 x + y}$
- Use it to compute $B(2.5, 1.5) - B(3.5, 0.5)$. (5 marks)
- (c) Find the general solution to the Bessel equation $x^2 y'' + xy' + (x^2 - 4)y = 0$. (5 marks)



QUESTION FOUR (20 MARKS)

4. (a) Use the Laplace transform table to find the inverse Laplace transform of the following:-
- (i) $F(S) = \frac{S+4}{S(S-1)(S^2+4)}$ (4 marks)
- (ii) $F(S) = \frac{S+2}{S^2-4S+13}$ (4 marks)
- (iii) $F(S) = \frac{18}{36S^2-24S+13}$ (4 marks)
- (b) Solve the Initial Value Problem $y'' + 9y = 6 \cos 3t$ for $y(0) = 2$ at $y'(0) = 0$. (5 marks)
- (c) Find the particular solution of the differential equation $y^4 y' = x + 1$ for $y(2) = 1$. (3 marks)

QUESTION FIVE (20 MARKS)

5. (a) With respect to the 2nd order linear differential equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$, define the following terms
- (i) An Analytic Function (2 marks)
- (ii) An Ordinary Point (2 marks)
- (iii) A regular-singular Point (2 marks)
- (b) Use Power Series method to solve the differential equation $y'' + x^2 y = 0$ near ordinary point $x_0 = 0$. (7 marks)
- (c) Show that $u = \frac{1}{\sqrt{x^2+y^2+z^2}}$ is a solution of the Laplace equation $\nabla^2 u = 0$. (7 marks)

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ii) $f(t) = t^{-1/2}$

(3 marks)

d) Determine the inverse Laplace transform of $F(s) = \frac{1}{s^2 - 5s + 6}$

(6 marks)



Show that $u = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$ is a solution of the Laplace equation

$\nabla^2 u = 0.$

(6 marks)

Question Two (20 marks)

2. a) State the Fourier series of a function $f(x)$. (4 marks).
- b) Given that $f(x) = x$ $0 < x < 2\pi$, use the series stated in (a) above to determine the Fourier series of (x) and sketch the graph from $x = -4\pi$ (10 marks)
- c) Differentiate between an odd and even function. State the properties of each function. Determine if the function $f(x) = x^3 \sin x$, is an odd or even function. (6 marks)

Question Three (20 marks)

3. a) i) Define the gamma function (3 marks)
- ii) Use the definition in a (i) above to evaluate the following integral.

$$\int_0^{\infty} \sqrt{y} e^{-y^3} dy + \int_0^{\infty} \sqrt{x} e^{-3\sqrt{x}} dx.$$

(6 marks)

b) Given that Beta function is defined as $B(x, y) = \frac{\int_0^x \sqrt{y}}{x+y}$

Compute:

- i) $B(2.5, 1.5)$ (3 marks)
- ii) $B(3.5, 0.5)$ (3 marks)

c) Given that $F(s) = \frac{1}{9s^2 + 6s + 1}$ determine $f(t)$. (5 marks)

QUESTION 5: (20 MARKS)

- (a) A material decays at a rate proportional to the amount present with a half life of 6 years. Determine the amount present after 5 years if the original mass was 120kg. (6 marks)
- (b) Use the method of undetermined coefficients to solve the differential equation $y''' - 5y'' + 25y' - 125y = 500x^2 - 1200x - 10$ (7 marks)
- (c) Use the variation of parameters method to solve the differential equation $y'' - 6y' + 9y = x^{-2}e^{3x}$ (7 marks)

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STION FIVE (20 MARKS)

The slope of a curve at any point is the reciprocal of twice the ordinate (y -value) of the point.

The curve also passes through the point $(4, 3)$. Find the equation of the curve.

(5 marks)

b) Solve $y'' - y' - 2y = (5x + 12)e^{2x}$

(8 marks)

c) Solve $y'' + y = \sec x$

(7 marks)

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2015/2016

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE (MECHANICAL ENGINEERING),

BACHELOR OF SCIENCE (CIVIL ENGINEERING) AND

BACHELOR OF SCIENCE (ELECTRICAL AND ELECTRONICS ENGINEERING)

ECU 203: ENGINEERING MATHEMATICS VIII

DATE: Thursday, 21st April 2016

TIME: 2.00 p.m. - 4.00 p.m.

INSTRUCTIONS:

Answer Question ONE and ANY OTHER TWO.

QUESTION ONE (COMPULSORY) (30 MARKS)

1. (a) Use Power Series method to solve the differential equation $y'' + x^2y = 0$ near ordinary point $x_0 = 0$. (6 marks)
- (b) (i) Define a Gamma function. (2 marks)
- (ii) Using the definition of the function in b(i) above, evaluate the integral.
$$\int_0^{\infty} \sqrt{x} e^{-3\sqrt{x}} dx$$
 (5 marks)
- (c) Given that $F(S) = \frac{13S-26}{6S^2-24S+78}$, find $F(t)$. (6 marks)
- (d) Use the Laplace Transform method to solve the differential equation
 $2y'' + 5y' + 2y = e^{-2t}$ for $y(0) = 1$, $y'(0) = 1$ (5 marks)
- (e) The periodic function of the voltage from a square wave generator is of the form
$$V(t) = \begin{cases} 0 & -4 < t < 0 \\ 10 & 0 < t < 4 \end{cases}$$

with a period of 8ms. Determine its Fourier Series. (6 marks)

QUESTION TWO (20 MARKS)

2. (a) Define the following terms with respect to the 2nd order linear differential equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$. (2 marks)
- (i) An Analytic Function (2 marks)
 - (ii) An Ordinary Point (2 marks)
 - (iii) A regular-singular point
- (b) Determine the recurrence relation of the differential equation $y'' - xy' - 2y = 0$ using Power Series near ordinary point $x_0 = 0$. (7 marks)
- (c) Determine a particular Polynomial solution to the Legendree equation $(1 - x^2)y'' - 2xy' + 30y = 0$. (Series solution not necessary) (7 marks)

QUESTION THREE (20 MARKS)

3. (a) Define and state the properties of odd and even functions. (6 marks)
- (b) (i) State the Fourier expansion series. (4 marks)
- (ii) Determine the Fourier series for the function defined by

$$f(x) = \begin{cases} 0 & \text{when } -2 < x < -1 \\ x & -1 < x < 1 \\ 5 & 1 < x < 2 \end{cases}$$

(10 marks)

QUESTION FOUR (20 MARKS)

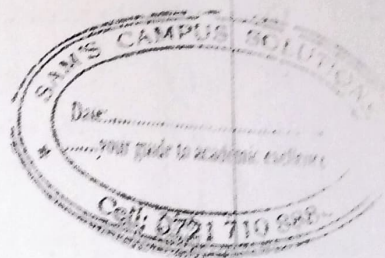
4. (a) Solve the differential equation $y' + y^2 + 1 = 0$ for $y(0) = 1$. (6 marks)
- (b) Find the general solution to the Bessel equation $x^2y'' + xy' + (x^2 - 4)y = 0$ (6marks)
- (c) Determine the particular solution of the differential equation $y^4y' = x + 1$ for $y(2) = 1$ (5 marks)
- (d) Use Laplace transform table to find the inverse Laplace transform of $F(S) = \frac{18}{36S^2 - 24S + 13}$ (3 marks)

ECU 203

21st APRIL 2016

QUESTION FIVE (20 MARKS)

5. (a) (i) Define a Laplace transform. (2 marks)
- (ii) Using the definition in 5(a)(i), determine the Laplace transform of $f(t) = t + e^{3t}$. (4 marks)
- (b) Find $f(t)$ given that
- (i) $F(S) = \frac{S+4}{S(S-1)(S^2+4)}$ (4 marks)
- (ii) $F(S) = \frac{S+2}{S^2-4S+13}$ (4 marks)
- (c) Use Laplace transform to solve the IVP $y'' + 9y = 6 \cos 3t$ for $y(0) = 2$ and $y'(0) = 0$. (6 marks)





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UNIVERSITY EXAMINATIONS 2014/2015

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
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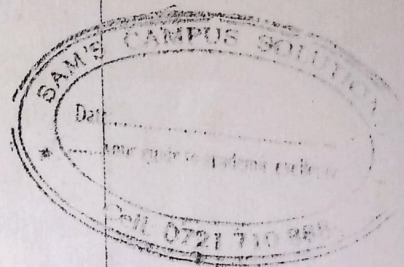
ECU 203: ENGINEERING MATHEMATICS VIII

DATE: TUESDAY 14TH APRIL 2015

TIME: 11.00 A.M. – 1.00 P.M.

INSTRUCTIONS

Answer Question ONE and ANY OTHER TWO



1. a) With respect to the 2nd order linear differential equation

$$a_2(x) y'' + a_1(x) y' + a_0(x) y = 0,$$

define the following terms

- i) Analytic functions
- ii) An Ordinary Point
- iii) A regular Singular Point

(2 marks)

(2 marks)

(2 marks)

- b) Determine the recurrence relation of the differential equation using power series near ordinary point $X_0 = 0$.

$$y'' - xy' - 2y = 0$$

(6 marks)

- c) i) Define a Laplace transform

(3 marks)

- ii) Determine the Laplace transform of $f(t) = e^{3t} + t$ using the definition stated in c(i) above.

(5 marks)

$$a_{n+1} (n+1)n x^n = a_n n x^n + 2a_n$$

$$a_{n+1} = \frac{a_n (n+2)}{n(n+1)}$$

- d) Find the Fourier series for the periodic function given that the voltage from a square wave generator is of the form

$$v(t) = \begin{cases} 0 & -4 < t < 0 \\ 10 & 0 < t < 4 \end{cases}$$

which has a period of 8 ms.

$$2s^2 + 4s + 5 + 2$$

$$2s(s+2) + 1(s+2)$$

$$= (2s+1)(s+2)$$

(5 marks)

- e) Solve the initial value problems using the Laplace Transform

$$2y'' + 5y' + 2y = e^{-2t} \text{ for } y(0) = 1 \text{ and } y'(0) = 1$$

(5 marks)

Question Two (20 marks)

- a) Determine a particular polynomial solution to the Legendree equation

$$(1-x^2) y'' - 2xy' + 30y = 0$$

(Series solutions not necessary)

(7 marks)

- b) Solve the differential equation

$$y'' + x^2 y = 0 \text{ using power series near ordinary point } X_0 = 0.$$

(7 marks)

- c) Find the general solution to the Bessel equation $x^2 y'' + xy' + x^2 - 4 = 0$

(6 marks)

Question Three (20 marks)

- a) Define the Fourier expansion series.

(4 marks)

- b) Determine the Fourier series for the function defined by

$$f(x) = \begin{cases} 0 & \text{where } -2 < x < -1 \\ 5 & \text{" } -1 < x < 1 \\ 0 & \text{" } 1 < x < 2 \end{cases}$$

(10 marks)

- c) Define the odd and even functions and state properties of each function.

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Question Four (20 Marks)

a) Find the inverse Laplace Transformation of the following

i) $F(S) = \frac{S+4}{S(S-1)(S^2+4)}$

ii) $F(S) = \frac{S+2}{S^2-4S+13}$

(5 marks)

b) Solve the initial value problem using Laplace method $y'' + 9y = 6 \cos 3t$
for $y(0)=2$ and $y'(0)=0$

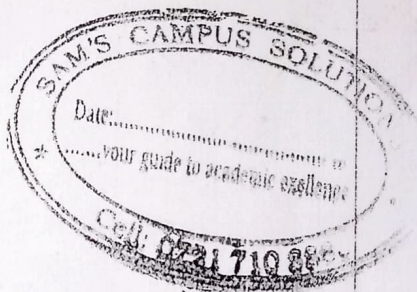
(10 marks)

Question Five (20 marks)

a) Solve the differential equation

$$\frac{dy}{dx} - y^2 - 1 = 0 \quad y(0) = 1$$

(use any convenient method)



(6 marks)

b) Determine the particular selection of the differential equation.

$$y^4 y' = x + 1 \quad \text{for } y(2) = 1$$

(6 marks)

c) State the particular polynomial solution to the Legendree equation

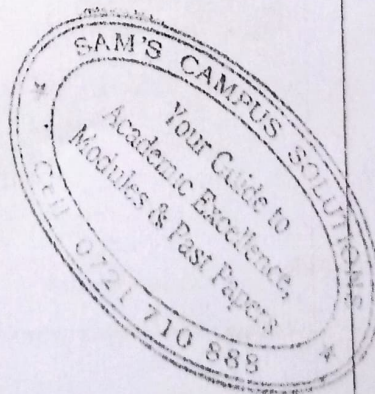
$$(1-x^2) y'' - 2xy' + 42y = 0$$

(8 marks)

21st APRIL 2016

QUESTION FIVE (20 MARKS)

5. (a) (i) Define a Laplace transform. (2 marks)
- (ii) Using the definition in 5(a)(i), determine the Laplace transform of $f(t) = t + e^{3t}$. (4 marks)
- (b) Find $f(t)$ given that
- (i) $F(S) = \frac{S+4}{S(S-1)(S^2+4)}$ (4 marks)
- (ii) $F(S) = \frac{S+2}{S^2-4S+13}$ (4 marks)
- (c) Use Laplace transform to solve the IVP $y'' + 9y = 6 \cos 3t$ for $y(0) = 2$ and $y'(0) = 0$. (6 marks)





KENYATTA UNIVERSITY

**EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENC
(AEROSPACE ENGINEERING), BACHELOR OF SCIENCE (AGRICULTURAL &
BIOSYSTEMS ENGINEERING), BACHELOR OF SCIENCE (CIVIL ENGINEERING),
BACHELOR OF SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING),
BACHELOR OF SCIENCE (MECHANICAL ENGINEERING) AND BACHELOR OF SCIENCE
(PETROLEUM ENGINEERING)**

ECU 203: ENGINEERING MATHEMATICS VIII

1ST SEMESTER 2021/2022

TIME: 2 HOURS

INSTRUCTIONS: Answer Question ONE and any other TWO Questions.

QUESTION ONE (30 MARKS)

- a) Given that $F(s) = \frac{s^2 - 26s - 47}{(s-1)(s+2)(s+5)}$, determine $f(t)$ (6 marks)
- b) Evaluate the integral: $\int_0^4 \frac{dt}{\sqrt{4t-t^2}}$ (5 marks)
- c) Obtain the value of $P_3(x)$ using the Rodrigues' formula (5 marks)
- d) Use power series to solve the differential equation: $y'' + xy' + y = 0$ (5 marks)
- e) Use Laplace transforms to solve the differential equation: $9y'' - 24y' + 16y = 0$, given that $y(0) = 3$ and $y'(0) = 3$ (9 marks)

QUESTION TWO (20 MARKS)

Evaluate the following integrals:

- a) $\int_0^\infty y^2 e^{-2y^2} dy$ (5 marks)
- b) $\int_0^2 \sqrt{x(2-x)} dx$ (5 marks)
- c) $\int_0^1 \left\{ \ln\left(\frac{1}{t}\right) \right\}^{-\frac{1}{2}} dt$ (5 marks)
- d) $\int_0^\infty u^{\frac{3}{2}} e^{-3u} du$ (5 marks)

QUESTION THREE (20 MARKS)

- a) Expand $f(x) = x^2, 0 < x < L$ in Fourier cosine series (4 marks)
- b) Show that $J_{\frac{5}{2}}(x) = \sqrt{\frac{2}{\pi x}} \left[\left(\frac{3-x^2}{x^2} \right) \sin x - \frac{3 \cos x}{x} \right]$ (6 marks)
- c) Use Laplace transforms to solve the differential equation $y'' - 2y' + y = 3e^{4t}$ given that $y(0) = -\frac{2}{3}$ and $y'(0) = \frac{13}{3}$ (10 marks)

QUESTION FOUR (20 MARKS)

- a) Evaluate $\int_0^{\frac{\pi}{2}} \sin^4 \theta \cos^5 \theta d\theta$ (3 marks)
- b) Determine the Laplace transform of the function $f(t) = 4t^2 - 3 \cos t + 5e^{-7t}$ (3 marks)
- c) Define the following terms:
- Periodic function (1 mark)
 - Half range Fourier series (1 mark)
- d) Obtain the Fourier series expansion of: $f(x) = \begin{cases} \pi - x, & 0 < x < \pi \\ 0, & \pi < x < 2\pi \end{cases}$ and has period 2π . Hence, deduce that $\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$ (12 marks)

QUESTION FIVE (20 MARKS)

- a) Given that $F(s) = \frac{1}{3-4s} + \frac{3-2s}{s^2+49}$, determine $f(t)$ (5 marks)
- b) Determine the particular polynomial solution to the Legendre's equation $(1-x^2)y'' - 2xy' + 42y = 0$ (6 marks)
- c) Show that $\sin \theta \frac{d^2y}{d\theta^2} + \cos \theta \frac{dy}{d\theta} + n(n+1) \sin \theta y = 0$ can be transformed into Legendre's equation by means of the substitution $x = \cos \theta$ (9 marks)

$$\begin{aligned} & \frac{9 + 25^2 - 8}{35-12} + \frac{17}{3} \\ & \frac{3(9 + 25^2 - 8) + 17(35-12)}{3(35-12)} \\ & = \frac{27 + 652 - 24 + 595 + 204}{95-36} \\ & = \frac{1529}{59} \end{aligned}$$

$$\begin{aligned} & \frac{9 + 25^2 - 8}{35-12} + \frac{17}{3} \\ & = \frac{9 + 25(35-4)}{3(35-12)} \\ & = \frac{9 + 25 \times 31}{3 \times 23} \\ & = \frac{9 + 775}{69} \\ & = \frac{784}{69} \end{aligned}$$

Question 1

1. Show that $J_{5/2} x = \sqrt{\frac{2}{\pi x}} \left(\frac{3-x^2}{x^2} \right) \sin x - \frac{3 \cos x}{x}$

$$J_{n+1}(x) = \frac{2n}{x} J_n(x) - J_{n-1}(x)$$

$$\therefore J_{(3/2+1)}(x) = \frac{2 \left(\frac{3}{2} \right)}{x} J_{3/2}(x) - J_{(3/2-1)}(x)$$

$$J_{3/2+1}(x) = \frac{3}{x} J_{3/2}(x) - J_{1/2}(x)$$

$$J_{3/2}(x) = J_{1/2+1}(x)$$

$$J_{1/2+1}(x) = \frac{2 \left(\frac{1}{2} \right)}{x} J_{1/2}(x) - J_{(-1/2)}(x)$$

$$\frac{1}{x} J_{1/2}(x) - J_{-1/2}(x)$$

$$= \frac{1}{x} \left(\sqrt{\frac{2}{\pi x}} \sin x \right) - \left(\sqrt{\frac{2}{\pi x}} \cos x \right) \Rightarrow \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right)$$

$$\cdot \frac{3}{x} \left(\sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right) - \frac{2}{\pi x} \sin x \right)$$

$$\sqrt{\frac{2}{\pi x}} \left(\frac{3 \sin x}{x^2} - \frac{3 \cos x}{x} \right) - \frac{2}{\pi x} \sin x$$

$$\sqrt{\frac{2}{\pi x}} \left(\frac{3 \sin x}{x^2} - \frac{3 \cos x}{x} - \sin x \right)$$

$$= \sqrt{\frac{2}{\pi x}} \left(\frac{3-x^2}{x^2} \right) \sin x - \frac{3 \cos x}{x} \quad \text{but } \frac{3}{x^2} - 1 = \frac{3-x^2}{x^2}$$

$$= \sqrt{\frac{2}{\pi x}} \left(\frac{3-x^2}{x^2} \right) \sin x - \frac{3 \cos x}{x}$$

Question 2.

Let $p = \cos \theta$ to show that $\sin \theta \frac{d^2 y}{dx^2} + \cos \theta \frac{dy}{dx} = n(n+1) \sin \theta y$ can be

transformed into a Legendre Equation.

The form of Legendre equation is:

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

$$p = \cos \theta$$

$$dp = -\sin \theta d\theta$$

$$\frac{dp}{d\theta} = -\sin \theta \quad \text{--- (i)}$$

$$\frac{dy}{d\theta} = \frac{dy}{dp} \cdot \frac{dp}{d\theta}$$

$$\frac{dy}{d\theta} = -\sin \theta \cdot \frac{dy}{dp} \quad \text{--- (ii)}$$

$$\frac{d^2 y}{d\theta^2} = -\cos \theta \cdot \frac{dy}{dp} - \sin \theta \cdot \frac{d^2 y}{dp^2} \cdot \frac{dp}{d\theta}$$

$$= -\cos \theta \cdot \frac{dy}{dp} - \sin \theta \cdot \frac{d^2 y}{dp^2} \cdot (-\sin \theta)$$

$$= -\cos \theta \frac{dy}{dp} + \sin^2 \theta \frac{d^2 y}{dp^2} \quad \text{--- (iii)}$$

$$\sin \theta \left[-\cos \theta \frac{dy}{dp} + \sin^2 \theta \frac{d^2 y}{dp^2} \right] + \cos \theta \left[-\sin \theta \frac{dy}{dp} \right] = n(n+1) \sin \theta y$$

Dividing all through by $\sin \theta$;

$$-\cos \theta \cdot \frac{dy}{dp} + \sin^2 \theta \cdot \frac{d^2 y}{dp^2} - \cos \theta \cdot \frac{dy}{dp} = n(n+1) y$$

$$\sin^2 \theta \frac{d^2 y}{dp^2} - 2\cos \theta \frac{dy}{dp} = -n(n+1) y$$

But $1 - \cos^2 \theta = \sin^2 \theta$

Continuation; Q2

$$(1 - \cos^2 \theta) \frac{d^2 y}{dp^2} - 2 \cos \theta \cdot \frac{dy}{dp} = -n(n+1)y$$

$$\text{Let } \frac{d^2 y}{dp^2} = y'', \quad \frac{dy}{dp} = y' \quad \text{and} \quad \cos \theta = p$$

$$(1 - p^2) y'' - 2 p y' = -n(n+1)y$$

$$(1 - p^2) y'' - 2 p y' + n(n+1)y = 0$$

Question 3.

Using special functions definitions, evaluate;

a) $\int_0^{\infty} y^{3/2} e^{-3y} dy.$

By definition; $\Gamma_n = \int_0^{\infty} y^{n-1} e^{-y} dy$

Let $3y = t$

$y = t/3$

$\frac{dy}{dt} = \frac{1}{3}$

$dy = \frac{1}{3} dt$

For $y=0$, $t=0$

$y=\infty$, $t=\infty$

$\int_0^{\infty} e^{-t} \left(\frac{t}{3}\right)^{3/2} \frac{1}{3} dt$

$\frac{1}{3^{5/2}} \int_0^{\infty} e^{-t} t^{3/2} dt$

$n-1 = 3/2$, $n = 5/2$

$= \frac{1}{3^{5/2}} \sqrt{\frac{5}{2}} = \frac{\frac{3}{4} \sqrt{\pi}}{(\sqrt{3})^5}$

b) $\int_0^{\infty} x^2 e^{-2x^2} dx$

$\Gamma_n = \int_0^{\infty} x^{n-1} e^{-x} dx$

Let $2x^2 = t$

$x^2 = t/2$

$x = \left(\frac{t}{2}\right)^{1/2}$ $\frac{dx}{dt} = \frac{\frac{1}{2} t^{-1/2}}{\sqrt{2}}$

$dx = \frac{\frac{1}{2} t^{-1/2}}{\sqrt{2}} dt$

For $x=0$, $t=0$

$x=\infty$, $t=\infty$

$$= \int_0^{\infty} \left(\frac{t}{2}\right) e^{-t} \frac{\frac{1}{2} t^{-\frac{1}{2}}}{\sqrt{2}} dt$$

$$= \frac{1}{2} \int_0^{\infty} e^{-t} t^{\frac{1}{2}} dt$$

$$n-1 = \frac{1}{2}, n = \frac{3}{2}$$

$$= \frac{\frac{1}{2}}{2\sqrt{2}} \Gamma_{\frac{3}{2}}$$

$$= \frac{\frac{1}{2}}{2\sqrt{2}} \cdot \frac{1}{2} \sqrt{\pi} = \underline{\underline{\frac{1}{8} \frac{\sqrt{\pi}}{\sqrt{2}}}}$$

$$(c) \int_0^{\infty} x^3 e^{-2x} dx$$

$$\Gamma_n = \int_0^{\infty} e^{-t} x^{n-1} dx$$

$$\text{Let } 2x = t$$

$$x = \frac{t}{2}$$

$$dx = \frac{1}{2} dt$$

$$\text{For } x=0, t=0$$

$$x=\infty, t=\infty$$

$$\int_0^{\infty} \left(\frac{t}{2}\right)^3 e^{-t} \frac{1}{2} dt$$

$$\frac{1}{2^4} \int_0^{\infty} t^3 e^{-t} dt$$

$$n-1=3, n=4$$

$$= \frac{1}{2^4} \Gamma_4$$

$$= \frac{1}{2^4} \cdot 3! = \underline{\underline{\frac{3}{8}}}$$

$$d) \int_0^3 (9-t)^{3/2} dt$$

$$\beta(m,n) = \int_0^1 t^{m-1} (1-t)^{n-1} dt = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$\text{Let } t^2 = 9p.$$

$$t = \sqrt{9p} = 3p^{1/2}$$

$$dt = \left(\frac{1}{2} \times 3 \times p^{-1/2}\right) dp$$

$$dt = \frac{3}{2} p^{-1/2} dp$$

$$\text{From; } \frac{t^2}{9} = p, \quad t=0, \quad p=0$$

$$t=3, \quad p=1$$

$$\beta(m,n) = \int_0^1 (9-9p)^{3/2} \cdot \frac{3}{2} p^{-1/2} dp$$

$$\beta(m,n) = \int_0^1 9^{3/2} (1-p)^{3/2} \cdot \frac{3}{2} p^{-1/2} dp$$

$$= 3^3 \cdot \frac{3}{2} \cdot \int_0^1 p^{-1/2} (1-p)^{3/2} dp$$

$$= 81 \cdot \frac{1}{2} \int_0^1 p^{-1/2} (1-p)^{3/2} dp$$

$$m-1 = -1/2, \quad m = 1/2$$

$$n-1 = 3/2, \quad n = 5/2$$

$$\beta(m,n) = \frac{81 \cdot \frac{1}{2} \times \frac{\Gamma(1/2) \Gamma(5/2)}{\Gamma(3)}}{\Gamma(3)}$$

$$= \frac{\sqrt{\pi} \times \frac{3}{4} \sqrt{\pi}}{2!} \times 81 \cdot \frac{1}{2}$$

$$\beta(m,n) = 81 \cdot \frac{3}{8} \pi = \frac{243}{16} \pi$$

$$(e) \int_0^4 \frac{dx}{\sqrt{4x-x^2}}$$

$$\text{Let } x^2 = 4xp.$$

$$x = 4p$$

$$dx = 4dp$$

$$\text{For } x=0, p=0$$

$$x=4, p=1$$

$$\beta(m,n) = \int_0^1 \frac{4 dp}{(16p-16p^2)^{\frac{1}{2}}}$$

$$= \int_0^1 4(16p-16p^2)^{-\frac{1}{2}} dp.$$

$$= \int_0^1 (4)(16p)^{-\frac{1}{2}} \cdot (1-p)^{-\frac{1}{2}} dp.$$

$$= 4 \cdot 16^{-\frac{1}{2}} \int_0^1 p^{-\frac{1}{2}} (1-p)^{-\frac{1}{2}} dp$$

$$= \int_0^1 p^{-\frac{1}{2}} (1-p)^{-\frac{1}{2}} dp.$$

$$m-1 = -\frac{1}{2}, m = \frac{1}{2}$$

$$n-1 = -\frac{1}{2}, n = \frac{1}{2}$$

$$\beta(m,n) = \frac{\Gamma\frac{1}{2} \cdot \Gamma\frac{1}{2}}{\Gamma\frac{1}{2} + \frac{1}{2}}$$

$$= \frac{\sqrt{\pi} \cdot \sqrt{\pi}}{\pi}$$

$$= \frac{\pi}{0!} = \underline{\underline{\pi}}$$

$$(f) \int_0^4 x^2 \sqrt{(16-x^2)} dx$$

$$\text{let } x^2 = 16p$$

$$x = \sqrt{16p} = 4p^{\frac{1}{2}}$$

$$dx = 2p^{-\frac{1}{2}} dp$$

$$\text{For } x=0, p=0$$

$$x=4, p=1$$

$$\beta(m,n) = \int_0^1 16p((16-16)p)^{\frac{1}{2}} 2p^{-\frac{1}{2}} dp$$

$$= \int_0^1 32 p^{1-\frac{1}{2}} (16-16p)^{\frac{1}{2}} dp$$

$$= \int_0^1 32 p^{\frac{1}{2}} \cdot 16^{\frac{1}{2}} (1-p)^{\frac{1}{2}} dp$$

$$= 32 \cdot 4 \int_0^1 p^{\frac{1}{2}} (1-p)^{\frac{1}{2}} dp$$

$$= 128 \int_0^1 p^{\frac{1}{2}} (1-p)^{\frac{1}{2}} dp$$

$$m-1 = \frac{1}{2}; m = \frac{3}{2}$$

$$n-1 = \frac{1}{2}; n = \frac{3}{2}$$

$$= 128 \times \frac{\sqrt{\frac{3}{2}} \cdot \sqrt{\frac{3}{2}}}{\sqrt{\frac{3}{2} + \frac{3}{2}}}$$

$$= 128 \cdot \frac{\frac{1}{2} \sqrt{\frac{1}{2}} \cdot \frac{1}{2} \sqrt{\frac{1}{2}}}{\sqrt{3}}$$

$$= \frac{128 \cdot \frac{1}{4} \cdot \sqrt{\pi} \cdot \sqrt{\pi}}{2!}$$

$$= 32 \times \pi \times \frac{1}{2}$$

$$= \underline{\underline{16\pi}}$$